

Code-Layout, Readability and Reusability

Date	2June2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5Marks

Code-Layout, Readability and Reusability

This document evaluates the quality of the EduGenie codebase in terms of its structure, readability, maintainability, and reusability. The project follows modular software engineering principles by separating the backend, frontend, AI services, and utility functions into independent components. Meaningful naming conventions, proper documentation, reusable functions, and consistent coding standards have been adopted throughout the project to improve code quality and simplify future enhancements.

CodeLayoutChecklist

S.No	CodeQuality Parameter	Description	Followed(Yes/ No/Partial)	Remarks
1	Consistent Indentation	Uniform indentation project.	and spacing Yes	ned throughout the PEP-coding standards follow
2	Proper File Structure	Backend, frontend, logically.	templates, staYes tic files, and AIModular FastAI	project structure.
3	Meaningful Variable Names	Variables use descriptive purpose.	ptive names representYes	ning their Easy to understand and maintain
4	Function / Method Names	Functions clearly describe their operations.	scribe their Yes	Self-explanatory API methods
5	Code Comments	Important logic is documented using docstrings.	ocumented using cYes	mmments anCritical modules documented.
6	Modular Design	Functionalities are divided into reusable modules.	divided into reusableYes	modules.
7	No Redundant Code	Duplicate code is minimized through reusable functions.	inimized through reusableYes	usable APIs.
8	Error Handling	API exceptions and runtime errors are gracefully.	runtime errors are Yes	handled Tryimplemented.

modules are org

AI services separated into

Helper functions reused a

Reusable Components/Modules

S.No	Component/ Module Name	Language/ Technology	Where Reused	Reusability Level
1	Question Answering Module	Python (Gemini API)	Educational Question Answering	High
2	Concept Explanation Module	Python (LaMini-Flan-T5)	Simplified Topic Explanation	High
3	Quiz Generation Module	Python (Gemini API)	Automatic MCQ Generation	High
4	Text Summarization Module	Python (Transformers)	Educational Content Summarization	High
5	Learning Path Recommendation Module	Python (Gemini API)	Personalized Study Roadmaps	High
6	FastAPI Backend	FastAPI	Handles API requests for all modules	High
7	HTML Templates	HTML + Jinja2	Shared user interface pages	Medium
8	CSS Stylesheets	CSS	Common styling across all pages	Medium
9	Utility Functions	Python	Reusable helper methods	High
10	Configuration Module	Python (.env)	API key and environment configuration	High

Overall Code Quality Assessment

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well-organized modular project architecture.

		Descriptive naming conventions and clean
Readability	5	
Reusability	5	AI modules and helper functions are reusable across
		multiple features.
Documentation/Comments	5	Important functions include comments and documentation.
Error Handling	5	Robust exception handling improves reliability.
OverallScore	5/5	Production-ready coding practices with scalable architecture.